ODD Protocol: Brock–Hommes Agent-Based Asset Pricing Model

Dominik Allen Minho Kang Saurav Jha

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Document scope

This ODD documents the implementation in adaptive-markets-lab/bh-agent-model. The repository supports three distinct operating modes, each documented below. Where the implementation contains internal differences, these are noted explicitly.

1 Overview

1.1 Purpose

The model studies how heterogeneous forecasting heuristics and performance-based evolutionary switching affect the relative dominance of trader types. It is implemented in three modes: (i) a *synthetic endogenous* mode that generates price paths entirely from internal dynamics; (ii) an *empirical replay* mode that uses real S&P 500 returns to drive fitness updates; and (iii) a *sensitivity analysis* mode that runs Sobol indices over both empirical returns and a 9-dimensional parameter space.

1.2 Entities, state variables, and scales

Trader types are instances of the `Trader` dataclass with fixed coefficients: trend-extrapolation g, constant bias b, per-period cost, and a name string. Two dynamic state variables are maintained per trader: `fitness` and `last_demand`. Factory defaults and the values used in the empirical script are:

Strategy	Default g	Default b	Default cost	Empirical script	Description
Fundamentalist	0.0	0.0	0.001	cost = 0.0002	Trades on fundamental value
Chartist	1.2	0.0	0.0	$g = 1.1$	Extrapolates past trend
Contrarian	-0.6	0.0	0.0	$g = -0.8$	Bets against current trend
Optimist	0.0	0.002	0.001	$b = 0.0005$, cost = 0.0001	Maintains constant bullishness

The `Market` class holds: `weights` (population shares), `x` (current price-deviation state), and fixed parameters β, r, σ^2, risk aversion, and noise standard deviation. One time step corresponds to one observation period. The model is non-spatial.

1.3 Process overview — three operating modes

Loop. At each time step, the model follows a consistent sequence: (1) Traders form forecasts based on past price deviations (2) Traders determine demand using a mean-variance rule (3) Market price is updated from aggregate demand (synthetic mode) or observed returns (empirical mode) (4) Strategy fitness is updated based on realized returns (5) Population weights are updated using a softmax rule.

Mode 1 (Synthetic). The repository contains a synthetic analysis module under `bh_strategy_simulation_synthetic_data`. This mode generates price paths from internal dynamics rather than external market data.

Mode 2 (Empirical replay). The script `bh_strategy_simulation_real_data.py` loads historical S&P 500 returns (`^GSPC`, 2016-01-01 to 2025-12-31) and runs an explicit loop. Each step: (1) computes trader demand from the previous observed return x_{t-1}; (2) updates fitness using the realized return signal; (3) applies softmax switching. The file instantiates traders with fundamentalist cost = 0.0002, chartist $g = 1.1$, contrarian $g = -0.8$, optimist $b = 0.0005$ and cost = 0.0001, and uses $\beta = 0.5$, $r = 1.01$, risk aversion = 5.0, and $\sigma^2 = \max(\hat{\sigma}^2_{\text{data}}, 10^{-4})$.

Mode 3 (Sensitivity). The script `bh_sensitivity_analysis.py` performs Sobol sensitivity analysis over a 9-parameter space and computes first-order (S_1) and total-order (S_T) indices for 9 output metrics. The code defines the Sobol problem directly in the file and documents total runs as $N(2k+2)$ for $k = 9$.

2 Design Concepts

Basic principles. The model is based on the Adaptive Belief System framework, which combines a standard asset pricing setup with an evolutionary selection mechanism. Instead of assuming fully rational agents, traders use simple rules and adapt over time by shifting toward strategies that perform well.

This approach connects ideas from behavioral economics, evolutionary game theory, and financial market modeling.

Objectives. Agents aim to maximize expected utility based on a mean-variance framework, balancing expected returns against risk.

Learning. Learning occurs through the fitness update mechanism, where strategies accumulate performance over time using an exponentially weighted average.

Emergence. In synthetic mode no macro-level behavior is imposed. Market dynamics such as bubbles, crashes, and regime switching emerge from the interaction of heterogeneous strategies and the feedback between prices, fitness, and population weights. In empirical mode, the price path is fixed; only market-share composition is emergent.

Adaptation. Coefficients g and b are fixed for each type. Adaptation is purely population-level: relative fitness changes determine how the weight vector evolves over time.

Prediction. Each trader forms a one-step-ahead forecast $f_h(x_{t-1}) = g_h x_{t-1} + b_h$. Fundamentalists predict mean reversion toward zero, chartists extrapolate trends, contrarians bet on reversals, optimists maintain a positive bias.

Sensing & interaction. Traders respond only to the shared market-level signal (x_{t-1} or the previous observed return). Interaction is indirect through competition for population share and, in synthetic mode, through their contribution to aggregate price formation.

Stochasticity. Gaussian noise $\varepsilon_t \sim \mathcal{N}(0, \texttt{noise_std}^2)$ enters price formation in synthetic mode. In empirical mode, the realized return sequence is deterministic once data are downloaded; randomness in the sensitivity analysis arises from the Saltelli parameter sampler, not from the simulation loop.

Observation. The empirical script records weight histories and plots the S&P 500 price series alongside per-trader weights. The Sobol script produces grouped bar charts of S_1 and S_T indices. The synthetic script produces four plots: a standard time series, a smoothed dominance comparison, a phase plot, and a regime-shaded overlay.

Collectives. Traders are grouped by strategy type, and the key collective variable is the population weight assigned to each strategy.

3 Details

3.1 Initialization

`Market.__post_init__()` sets weights uniformly across all trader types, $x = 0$, and resets all trader state to zero. In the empirical script, weights are initialized uniformly and traders are explicitly reset before the loop. In `bh_strategy_simulation_base.py`, $x_0 = 0.10$ and the simulation uses `Market` for parameter validation and uniform weight initialization, then manages state explicitly in the loop.

3.2 Input data

Empirical mode and the Sobol script require `yfinance` data: adjusted closing prices, aligned dates, log returns, and return variance for `^GSPC` over 2016-01-01 to 2025-12-31. Synthetic mode and the bifurcation experiments require no external data.

3.3 Submodels

Forecasting.

$$f_h = g_h\, x_{t-1} + b_h.$$

Demand.

$$z_h = \frac{f_h - r\, x_{t-1}}{a\, \sigma^2},$$

where a is the common risk-aversion parameter and σ^2 is a constant variance estimate. Each call to `trader.demand()` stores z_h in `last_demand`.

Fitness update (memory parameter $\eta = 0.5$, hard-coded):

$$\pi_{h,t} = \text{realized_return}_t \times z_{h,t-1}, \qquad U_{h,t} = \eta\, U_{h,t-1} + (1-\eta)\, \pi_{h,t} - C_h.$$

In empirical mode, `realized_return` $= r_t$ (the observed return at time t, one step ahead of the demand calculation at $t-1$). In synthetic mode, `realized_return` $= x_t - x_{t-1}$.

Softmax switching (numerically stabilized):

$$n_{h,t+1} = \frac{\exp\!\big(\beta(U_{h,t} - \max_k U_{k,t})\big)}{\sum_j \exp\!\big(\beta(U_{j,t} - \max_k U_{k,t})\big)}.$$

`Market.softmax()` additionally clips scaled values to $[-500, 0]$ and falls back to uniform weights if the sum is zero or non-finite.

Price formation — synthetic mode (`Market.step()`):

$$x_t = \frac{\sum_h n_{h,t}\, z_{h,t} + \varepsilon_t}{r}, \qquad \varepsilon_t \sim \mathcal{N}(0, \texttt{noise_std}^2).$$

Price is formed from *weighted demands* $z_{h,t}$.

3.4 Sobol sensitivity parameters and outputs

Nine parameters are varied: $\beta \in [0, 5]$, $r \in [1.001, 1.05]$, $\sigma^2 \in [10^{-5}, 10^{-2}]$, risk aversion $\in [1, 20]$, $g_{\text{chartist}} \in [0.5, 2]$, $g_{\text{contrarian}} \in [-2, -0.1]$, $b_{\text{optimist}} \in [0, 0.005]$, $c_{\text{fundamentalist}} \in [0, 0.002]$, $c_{\text{optimist}} \in [0, 0.002]$. Nine output metrics are computed: mean weight and dominance fraction for each of the four trader types, plus overall switching volatility (mean standard deviation of weights across periods).

3.5 Model differences across operating modes

The three operating modes differ in how price dynamics and parameter values are specified.

In the synthetic mode, price is formed endogenously from the aggregation of trader demands, with an added stochastic noise term. This mode uses a high intensity-of-choice parameter ($\beta = 200$) and a fixed variance ($\sigma^2 = 0.25$), calibrated for numerical stability and to generate rich internal dynamics.

In the empirical replay mode, price dynamics are exogenously determined by observed S&P 500 returns. Strategy fitness is updated using realized returns, but prices themselves are not influenced by trader demand. This mode uses a lower intensity-of-choice parameter ($\beta = 0.5$) and a variance estimated from the data ($\sigma^2 \approx 10^{-4}$).

As a result, the two modes produce qualitatively different dynamics. The synthetic mode generates endogenous market behavior, including regime switching and volatility patterns, while the empirical mode isolates the evolution of strategy composition under real market conditions.

The sensitivity analysis extends the empirical framework by varying nine key parameters and evaluating their influence on model outputs using Sobol indices.